

Appendix deleted

1 Numerical Hamiltonians and one-sided derivative reconstruction

For the monotone theory in Section 3, the numerical Hamiltonian $\hat{H}(a, b)$ is assumed to be consistent with $-p^2$, nondecreasing in its first argument, and nonincreasing in its second argument. Two standard choices are the Lax–Friedrichs Hamiltonian

$$\hat{H}_{\text{LF}}(a, b) = -\left(\frac{a+b}{2}\right)^2 + \frac{\alpha}{2}(a-b), \quad (1.1)$$

where α is chosen no smaller than the relevant characteristic speed, and the Godunov Hamiltonian

$$\hat{H}_G(a, b) = -(a^-)^2 - (b^+)^2, \quad a^- := \min\{a, 0\}, \quad b^+ := \max\{b, 0\}. \quad (1.2)$$

The stationary phase estimate in Proposition ?? uses $\hat{H}_G$.

Some numerical runs use a high-order one-sided WENO reconstruction for the derivatives entering the Hamiltonian. This is an implementation option, not the monotone Hamiltonian used in the proof. For a periodic grid function u_k , define the left-biased quantities

$$\begin{aligned} a &= u_{k-3} - 2u_{k-2} + u_{k-1}, & b &= u_{k-2} - 2u_{k-1} + u_k, \\ c &= u_{k-1} - 2u_k + u_{k+1}, & d &= u_k - 2u_{k+1} + u_{k+2}. \end{aligned}$$

Set

$$\begin{aligned} I_0 &= 13(a-b)^2 + 3(a-3b)^2, & I_1 &= 13(b-c)^2 + 3(b+c)^2, \\ I_2 &= 13(c-d)^2 + 3(3c-d)^2, \end{aligned}$$

and

$$\alpha_0 = \frac{1}{(\epsilon_0 + I_0)^2}, \quad \alpha_1 = \frac{6}{(\epsilon_0 + I_1)^2}, \quad \alpha_2 = \frac{3}{(\epsilon_0 + I_2)^2}, \quad \omega_r = \frac{\alpha_r}{\alpha_0 + \alpha_1 + \alpha_2}.$$

The left derivative reconstruction used in the implementation is

$$\begin{aligned} p_k^- &= \frac{1}{\Delta\theta} \left[\frac{-(u_{k-1} - u_{k-2}) + 7(u_k - u_{k-1}) + 7(u_{k+1} - u_k) - (u_{k+2} - u_{k+1})}{12} \right. \\ &\quad \left. - \frac{\omega_0}{3}(a - 2b + c) - \frac{\omega_2 - 1/2}{6}(b - 2c + d) \right]. \end{aligned} \quad (1.3)$$

The right-biased derivative p_k^+ is obtained by applying the same formula to the reversed stencil. Equivalently, with

$$a^+ = u_{k+3} - 2u_{k+2} + u_{k+1}, \quad b^+ = u_{k+2} - 2u_{k+1} + u_k,$$

$$c^+ = u_{k+1} - 2u_k + u_{k-1}, \quad d^+ = u_k - 2u_{k-1} + u_{k-2},$$

and the corresponding weights ω_0^+, ω_2^+ ,

$$p_k^+ = \frac{1}{\Delta\theta} \left[\frac{-(u_{k-1} - u_{k-2}) + 7(u_k - u_{k-1}) + 7(u_{k+1} - u_k) - (u_{k+2} - u_{k+1})}{12} + \frac{\omega_0^+}{3}(a^+ - 2b^+ + c^+) + \frac{\omega_2^+ - 1/2}{6}(b^+ - 2c^+ + d^+) \right]. \quad (1.4)$$

The WENO option then uses a Roe-type entropy-fix numerical Hamiltonian for $H(p) = -p^2$ with the reconstructed values (p_k^+, p_k^-) . Since this high-order reconstruction is not monotone in the sense assumed in Section 3, the numerical experiments report it separately from the monotone Godunov/Lax–Friedrichs choices when needed.

2 Conservative fluxes for the split amplitude equation

The continuous conservative flux in the split amplitude equation is

$$F = -W \partial_\theta u.$$

On a uniform periodic trait grid, write

$$D_\theta \hat{F}(W, u)_{j,k} = \frac{\hat{F}_{j,k+1/2} - \hat{F}_{j,k-1/2}}{\Delta\theta}, \quad a_{k+1/2} = -\frac{u_{k+1} - u_k}{\Delta\theta}.$$

The upwind flux is

$$\hat{F}_{j,k+1/2}^{\text{up}} = a_{k+1/2}^+ W_{j,k} + a_{k+1/2}^- W_{j,k+1}, \quad a^+ = \max\{a, 0\}, \quad a^- = \min\{a, 0\}. \quad (2.5)$$

A Lax–Friedrichs flux is

$$\hat{F}_{j,k+1/2}^{\text{LF}} = \frac{1}{2} a_{k+1/2} (W_{j,k} + W_{j,k+1}) - \frac{1}{2} \alpha_{k+1/2} (W_{j,k+1} - W_{j,k}), \quad \alpha_{k+1/2} \geq |a_{k+1/2}|. \quad (2.6)$$

Both fluxes are conservative in the periodic trait variable. The upwind flux also satisfies the quasi-positivity condition used in Lemma ??.

3 Scalar mass-balance correction

A detailed algebraic form of the scalar mass-balance correction used in the fully discrete method is provided here. All spatial integrals below are understood in the discrete sense and may be evaluated by any consistent spatial quadrature used in the fully discrete scheme. The correction is based on the mass identity

$$\varepsilon \frac{d}{dt} \int_\Omega \rho(x, t) dx = \int_\Omega \rho(x, t) (K(x) - \rho(x, t)) dx.$$

Let $\widetilde{W}^{n+1}$, $\widetilde{u}^{n+1}$, and $\widetilde{\rho}^{n+1}$ denote the provisional amplitude, phase, and reconstructed spatial marginal after one time step. The correction is defined by

$$W^{n+1} = \lambda \widetilde{W}^{n+1}, \quad u^{n+1} = \widetilde{u}^{n+1}, \quad \lambda > 0.$$

It follows that

$$\rho^{n+1} = \lambda \widetilde{\rho}^{n+1}, \quad n^{n+1} = \lambda \widetilde{n}^{n+1}, \quad P^{n+1} = \lambda \widetilde{P}^{n+1}.$$

Thus the correction changes only the density scale and leaves the phase profile unchanged.

For compactness, define

$$\widetilde{M} = \int_{\Omega} \widetilde{\rho}^{n+1} dx, \quad \widetilde{S} = \int_{\Omega} K \widetilde{\rho}^{n+1} dx, \quad \widetilde{Q} = \int_{\Omega} (\widetilde{\rho}^{n+1})^2 dx,$$

and

$$M^n = \int_{\Omega} \rho^n dx, \quad R^n = \int_{\Omega} \rho^n (K - \rho^n) dx.$$

If the trapezoidal discretization is applied in time, the correction factor λ is determined by

$$\varepsilon \frac{\lambda \widetilde{M} - M^n}{\tau} = \frac{1}{2} [R^n + \lambda \widetilde{S} - \lambda^2 \widetilde{Q}].$$

Equivalently, λ is the positive root of

$$\mathcal{A}_{\text{TR}} \lambda^2 + \mathcal{B}_{\text{TR}} \lambda + \mathcal{C}_{\text{TR}} = 0,$$

where

$$\mathcal{A}_{\text{TR}} = \frac{\tau}{2} \widetilde{Q}, \quad \mathcal{B}_{\text{TR}} = \varepsilon \widetilde{M} - \frac{\tau}{2} \widetilde{S}, \quad \mathcal{C}_{\text{TR}} = -\varepsilon M^n - \frac{\tau}{2} R^n.$$

If a backward Euler discretization of the reaction term is applied, λ is determined by

$$\varepsilon \frac{\lambda \widetilde{M} - M^n}{\tau} = \lambda \widetilde{S} - \lambda^2 \widetilde{Q},$$

or equivalently

$$\mathcal{A}_{\text{BE}} \lambda^2 + \mathcal{B}_{\text{BE}} \lambda + \mathcal{C}_{\text{BE}} = 0,$$

with

$$\mathcal{A}_{\text{BE}} = \tau \widetilde{Q}, \quad \mathcal{B}_{\text{BE}} = \varepsilon \widetilde{M} - \tau \widetilde{S}, \quad \mathcal{C}_{\text{BE}} = -\varepsilon M^n.$$

The positive root is selected in both cases to preserve positivity of the reconstructed density.

When the provisional reconstruction already satisfies the discrete mass balance, the correction factor is close to one. In the small mutation regime, however, a small mass inconsistency may lead to a visible error in the peak height of the trait marginal. The scalar correction therefore acts as a mass-balance projection for the reconstructed density-level quantities. It improves the stability of the total mass and the peak height without changing the phase-based peak location.

4 Density and trait-marginal reconstruction, gauge normalization, and dual trait grids

The reconstruction operators used in the code act on the WKB variables and do not solve the original density equation. The default safe reconstruction of ρ is the log-stabilized quadrature. With

$$\ell_{j,k} = \log(\max\{W_{j,k}, W_{\text{floor}}\}) + \frac{u_k}{\varepsilon},$$

the code evaluates

$$\rho_j = \sum_k \omega_k \exp(\ell_{j,k})$$

by a row-wise log-sum-exp formula. This is algebraically identical to the composite quadrature

$$\rho_{j,Q} = \Delta\theta \sum_k W_{j,k} \exp(u_k/\varepsilon),$$

but it avoids underflow and overflow when ε is small. The max-gauge (??) is applied after each WKB step. It leaves $n_{j,k} = W_{j,k} \exp(u_k/\varepsilon)$ unchanged and keeps $\max_k u_k = 0$.

A Laplace reconstruction can be used near a nondegenerate phase maximizer. Let k_* be a nodal maximizer of u_k . The periodic stencil is represented in local coordinates

$$z_{k_*-1} = -\Delta\theta, \quad z_{k_*} = 0, \quad z_{k_*+1} = \Delta\theta,$$

even when the stencil crosses the endpoint of $[0, 1)$. With

$$d_{\theta\theta}u_{k_*} = \frac{u_{k_*+1} - 2u_{k_*} + u_{k_*-1}}{(\Delta\theta)^2},$$

the parabolic vertex is

$$z_* = -\frac{\Delta\theta}{2} \frac{u_{k_*+1} - u_{k_*-1}}{u_{k_*+1} - 2u_{k_*} + u_{k_*-1}}. \quad (4.7)$$

The reconstructed trait location is $\theta_* = (\theta_{k_*} + z_*) \bmod 1$. Interpolating W and u quadratically on this local stencil gives $W(x_j, \theta_*)$ and $u(\theta_*)$, and

$$\rho_j^{\text{Lap}} = W(x_j, \theta_*) \exp\left(\frac{u(\theta_*)}{\varepsilon}\right) \left(\frac{2\pi\varepsilon}{|d_{\theta\theta}u_{k_*}|}\right)^{1/2}. \quad (4.8)$$

In the implementation this expression is evaluated in logarithmic form. If $d_{\theta\theta}u_{k_*} \geq 0$, the curvature is too small, or the interpolated amplitude is non-positive, the hybrid reconstruction falls back to the log-stabilized direct quadrature.

The trait marginal is reconstructed differently from ρ . For WKB data, the code first forms the amplitude marginal

$$A_k = \int_{\Omega} W(x, \theta_k) dx$$

and then works with

$$L_k = \log P_k = \frac{u_k}{\varepsilon} + \log(\max\{A_k, A_{\text{floor}}\}). \quad (4.9)$$

Thus interpolation is applied to $L = \log P$, not to P itself. The peak search uses a periodic WENO interpolation of L_k on a fine uniform search grid, followed by the three-point parabolic refinement (??) applied to the reconstructed values on that search grid. This is why the reported small- ε peak location is less sensitive to whether the true density peak lies between two coarse trait nodes.

After the peak location θ_* has been found, the code can reconstruct a local peak shape. In the phase-amplitude form, it fits

$$u(\theta_* + z) \simeq \alpha_0 + \alpha_1 z + \alpha_2 z^2, \quad \log A(\theta_* + z) \simeq \beta_0 + \beta_1 z + \beta_2 z^2$$

on a small periodic stencil, optionally with weights that emphasize points near the peak. Hence

$$\log P(\theta_* + z) \simeq \left(\frac{\alpha_0}{\varepsilon} + \beta_0\right) + \left(\frac{\alpha_1}{\varepsilon} + \beta_1\right) z + \left(\frac{\alpha_2}{\varepsilon} + \beta_2\right) z^2.$$

When the quadratic coefficient is negative, the fitted peak width is

$$\sigma_P \simeq \frac{1}{\sqrt{-2(\alpha_2/\varepsilon + \beta_2)}}. \quad (4.10)$$

The reconstructed shape is then normalized by a prescribed mass,

$$P_{\text{rec}}(\theta) = M_{\text{target}} \frac{\exp(\widehat{L}(\theta) - \max_{\eta} \widehat{L}(\eta))}{\int_{\mathbb{T}} \exp(\widehat{L}(\eta) - \max_{\xi} \widehat{L}(\xi)) d\eta}, \quad (4.11)$$

where M_{target} is the solver-corrected mass when such a correction is available and otherwise the raw log-quadrature mass. This mass normalization is what stabilizes the reconstructed peak height after the location and width have been determined.

For the dual trait-grid implementation, let Θ_u be the fine phase grid and Θ_W the coarse amplitude grid. The reconstruction first computes

$$W_U = I_{W \rightarrow u} W_c,$$

where $I_{W \rightarrow u}$ is a periodic interpolation operator, and then applies the log-stabilized quadrature on Θ_u :

$$\rho_j = \sum_{\theta_k \in \Theta_u} \omega_k \exp \left(\log(\max\{(W_U)_{j,k}, W_{\text{floor}}\}) + \frac{u_k}{\varepsilon} \right).$$

The phase equation and the peak localization are therefore resolved on the fine one-dimensional grid, whereas the space-dependent amplitude is stored and advanced on the coarse grid. Since W carries all spatial degrees of freedom, this reduction is especially useful when the spatial discretization is large or high-dimensional.

5 Density-compatible variant and a sufficient condition for the one-sided remainder bound

We record a density-compatible variant of the semi-discrete WKB formulation. This variant is not needed in the statement of the main scheme, but it gives a concrete setting in which the weighted remainder condition in Assumption ?? can be verified by discrete summation by parts.

The phase equation is kept unchanged. The amplitude equation is replaced by

$$\begin{aligned} \varepsilon \frac{d}{dt} W_{j,k}^\varepsilon - D_k \delta_x^2 W_{j,k}^\varepsilon - \varepsilon^2 (E_k^\varepsilon)^{-1} \delta_\theta^2 (W_{j,\cdot}^\varepsilon, E_\cdot^\varepsilon)_k \\ = W_{j,k}^\varepsilon (K_j - \rho_{j,Q}^\varepsilon + \mathcal{H}_k(\rho_Q) + \widehat{H}(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon) - \varepsilon \delta_\theta^2 u_k^\varepsilon), \end{aligned} \quad (5.12)$$

where

$$\delta_\theta^2 (W_{j,\cdot}^\varepsilon, E_\cdot^\varepsilon)_k = \frac{W_{j,k+1}^\varepsilon E_{k+1}^\varepsilon - 2W_{j,k}^\varepsilon E_k^\varepsilon + W_{j,k-1}^\varepsilon E_{k-1}^\varepsilon}{(\Delta\theta)^2}.$$

Multiplying (??) by E_k^ε and using the phase equation gives

$$\varepsilon \frac{d}{dt} n_{j,k}^\varepsilon - D_k \delta_x^2 n_{j,k}^\varepsilon - \varepsilon^2 \delta_\theta^2 n_{j,k}^\varepsilon = (K_j - \rho_{j,Q}^\varepsilon) n_{j,k}^\varepsilon. \quad (5.13)$$

Thus the reconstructed density satisfies a conservative semi-discrete analogue of the original density equation. In the notation of (??), the remainder is exactly

$$R_{j,k}^\varepsilon = \varepsilon^2 \delta_\theta^2 n_{j,k}^\varepsilon.$$

Consequently, the weighted remainder satisfies

$$R_j^{\varepsilon,D} = \varepsilon^2 \Delta\theta \sum_k \frac{\delta_\theta^2 n_{j,k}^\varepsilon}{D_k}.$$

Using periodic summation by parts in the trait variable, we obtain

$$R_j^{\varepsilon,D} = \varepsilon^2 \Delta\theta \sum_k n_{j,k}^\varepsilon \delta_\theta^2 \left(\frac{1}{D_k} \right).$$

Hence, for $0 < \varepsilon \leq 1$,

$$(R_j^{\varepsilon,D})_+ \leq C_{D,h} m_j^\varepsilon, \quad C_{D,h} = \max_k \left| \delta_\theta^2 \left(\frac{1}{D_k} \right) \right|,$$

where

$$m_j^\varepsilon = \Delta\theta \sum_k n_{j,k}^\varepsilon.$$

This verifies Assumption ?? with $C_R = C_{D,h}$ and $r_h = 0$. If D is smooth and a standard centered periodic difference is used, $C_{D,h}$ remains bounded under regular trait-grid refinement.

6 Direct density solver and two-dimensional finite-element realization

For comparison, the original density equation can be discretized by the old linear implicit update

$$\varepsilon \frac{n_{j,k}^{n+1} - n_{j,k}^n}{\tau} - D_k \delta_x^2 n_{j,k}^{n+1} - \varepsilon^2 \delta_\theta^2 n_{j,k}^{n+1} = n_{j,k}^{n+1} (K_j - \rho_j^n), \quad \rho_j^n = \Delta\theta \sum_k n_{j,k}^n. \quad (6.14)$$

In the revised code, the direct density solver also accepts the Patankar reaction option

$$\varepsilon \frac{n_{j,k}^{n+1} - n_{j,k}^n}{\tau} - D_k \delta_x^2 n_{j,k}^{n+1} - \varepsilon^2 \delta_\theta^2 n_{j,k}^{n+1} = (K_j - \rho_j^n)_+ n_{j,k}^n - (K_j - \rho_j^n)_- n_{j,k}^{n+1}. \quad (6.15)$$

This option is used to match the positivity-oriented WKB update when direct and WKB runs are compared. The same spatial and trait grids are used for the direct and WKB solvers in the one-dimensional comparison tests.

For the two-dimensional spatial experiment, the spatial Laplacian and the eigenvalue problem are discretized by linear finite elements on a triangular mesh. For each trait node θ_k , the discrete effective Hamiltonian is computed from the generalized eigenvalue problem

$$A_k(\rho_Q) N^{(k)} = H_k(\rho_Q) M N^{(k)}, \quad (6.16)$$

where M is the finite-element mass matrix and

$$(A_k)_{ab} = D_k \int_\Omega \nabla \varphi_a \cdot \nabla \varphi_b \, dx - \int_\Omega (K(x) - \rho_Q(x)) \varphi_a \varphi_b \, dx. \quad (6.17)$$

The Neumann boundary condition is natural in this finite-element formulation. The phase equation remains one-dimensional in θ , and the amplitude equation is solved for all spatial finite-element degrees of freedom and trait nodes.